STA 5635 HW 6

Palmer Swanson

February 14, 2026

Problem 1

(1a) Two random forest classifiers. One with 22 features, the other with 500. Table in figure 1

Model	Train Misclass. Error	Test Misclass. Error
Random Forest, 22 Features	0.000000	0.280000
Random Forest, 500 Features	0.000000	0.156667

Table 1: Train and Test Misclassification Errors, part a

(1b) Compute feature importances using mean decrease in impurity and the permutation feature importances.

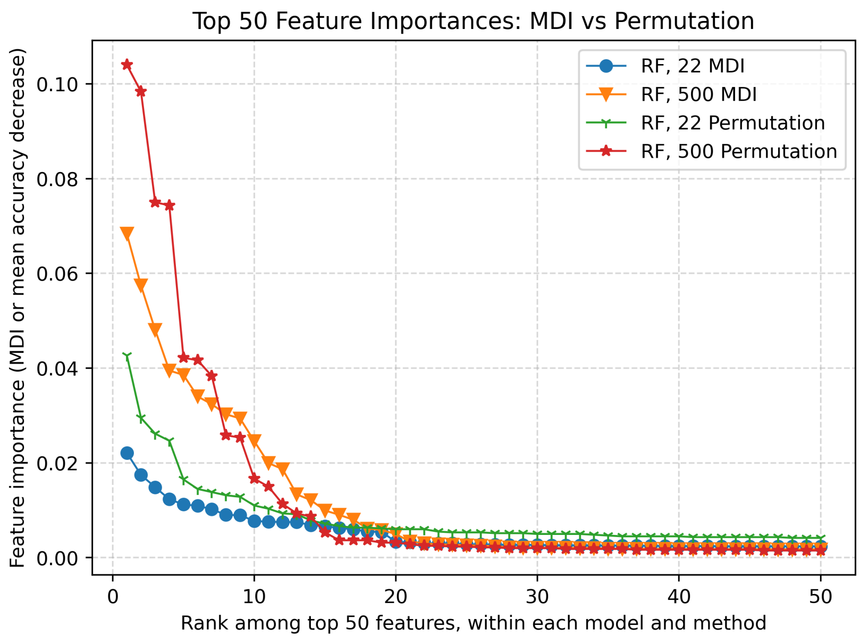

Figure 1: Feature Importances for the two RF, computed via MDI and via Permutations, part b.

(1c) Selecting k most important features.

Label	k	Train Misclass. Error	Test Misclass. Error
RF_22_MDI	10	0.000000	0.148333
RF_22_MDI	20	0.000000	0.120000
RF_22_MDI	30	0.000000	0.125000
RF_22_MDI	40	0.000000	0.128333
RF_500_MDI	10	0.000000	0.126667
RF_500_MDI	20	0.000000	0.115000
RF_500_MDI	30	0.000000	0.138333
RF_500_MDI	40	0.000000	0.148333
RF_22_PERM	10	0.000000	0.120000
RF_22_PERM	20	0.000000	0.126667
RF_22_PERM	30	0.000000	0.133333
RF_22_PERM	40	0.000000	0.141667
RF_500_PERM	10	0.000000	0.120000
RF_500_PERM	20	0.000000	0.135000
RF_500_PERM	30	0.000000	0.123333
RF_500_PERM	40	0.000000	0.141667

Table 2: Training and test misclassification errors for the fitted models, part c.

(1d) asdf

(1e) Of the models,

Code